WASP-43b

R-0460

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_240203` · created 2026-07-19T20:13:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460343.8176 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.058
Transit depth (Rp/Rs)^2	3.93 +/- 2.28 [%]
Orbital Inclination (inc)	86.79 +/- 2.92
Airmass coefficient 1 (a1)	2.72 +/- 1.38
Airmass coefficient 2 (a2)	-0.74 +/- 0.66
Scatter in the residuals of the lightcurve fit is	46.98 %
Best Comparison Star	#3 - [302, 370]
Optimal Aperture	2.45
Optimal Annulus	9.32
Transit Duration (day)	0.0611 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.058 vs 0.1594 (deviation 0.67σ)
Duration fit vs published	0.0611 d vs 0.0512 d (deviation 1.82σ)
Mid-time O–C	-5.9 min
Depth SNR	3.4
Residual scatter	46.98 %

Light curve

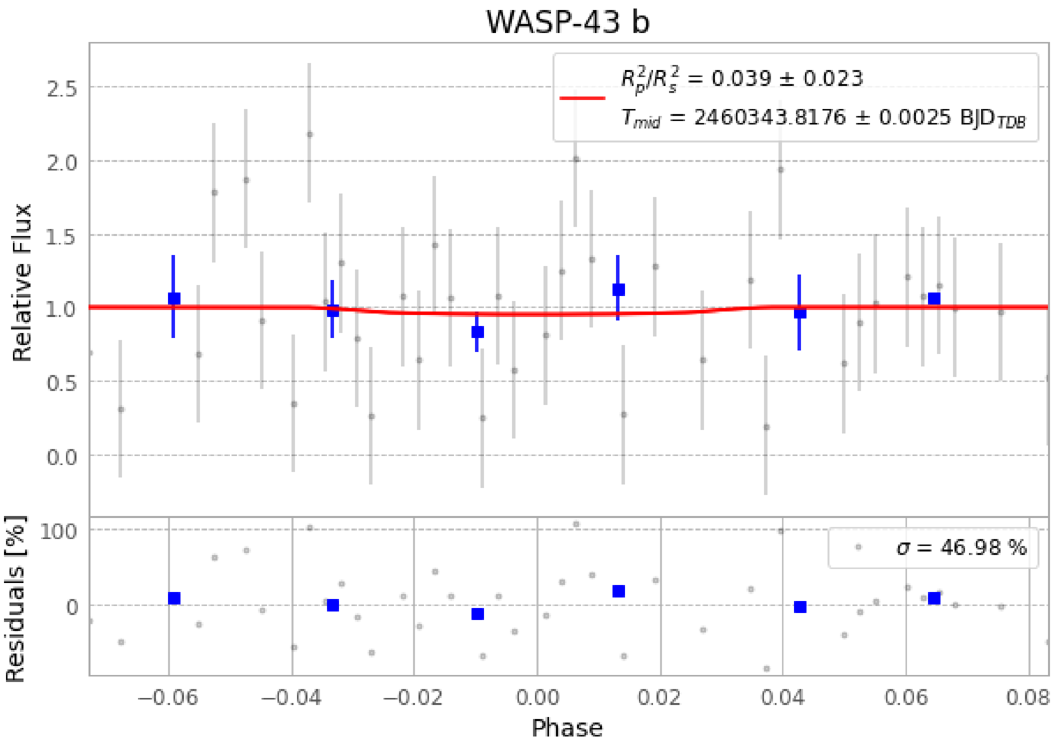

Figure 1 — FinalLightCurve_WASP-43 b_2024-02-02.png (SHA-256 7344f99e64dafe48…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [580, 457], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e23b9b093447f482…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

64 FITS frames (42 MB), `WASP-43240203060315.FITS` → `WASP-43240203091215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240203.log (1d04e4b1b65b…), FinalLightCurve_WASP-43 b_2024-02-02.png (7344f99e64da…), FinalLightCurve_WASP-43 b_2024-02-02.csv (0f3f5f7241ec…)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-19 20:13Z.